

OSMOSIS & THE SHRINKING POTATO

Interactive Food Science & Biology Laboratory Investigation

ELEMENTARY SCIENCE LAB

Student Name: _____

Class/Sec: _____

Date: _____

1. KITCHEN STORY & CORE SCIENTIFIC THEORY

Kabir's Kitchen Story & Observation

Kabir noticed that fresh cucumber slices became droopy and soft when sprinkled with salt! Why does salt water change the texture, mass, and length of fresh vegetables? Let's conduct a quantitative laboratory experiment using potato strips, salt solutions, rulers, and balances to find out!

What is Osmosis?

Osmosis is the movement of water molecules through a semi-permeable membrane from an area of *higher water concentration* (lower salt) to an area of *lower water concentration* (higher salt).

- **Semi-permeable Membrane:** A microscopic barrier (like a cell wall) that lets tiny water molecules pass through while blocking larger salt particles.
- **Turgor Pressure:** Water pushing outward against plant cell walls, making them firm and crisp (turgid).

3 Solution Conditions:

- **HYPOTONIC Solution A (Pure Water):** Water concentration outside is higher than inside. Water enters potato cells → gain mass, stiff/turgid texture.
- **ISOTONIC Solution B (Light Salt):** Equal water concentration inside and outside. Minimal net water flow.
- **HYPERTONIC Solution C (Concentrated Salt):** Salt water outside steals water from potato cells → lose mass, soft/limp/flaccid texture.

-  **Osmosis in Everyday Life:**
- **Raw Mango Pickle (Achar):** Salt draws water out of mangoes via osmosis to prevent spoilage.
 - **Soaking Plump Raisins:** Dry raisins swell up when placed in pure water as water diffuses inside.
 - **Crisp Lettuce Trick:** Sprinkling water on wilted lettuce makes it crisp again as water rushes into plant cells!

2. DETAILED EXPERIMENTAL SETUP & VARIABLE GUIDE

Setup Variable	Solution A (Pure Water)	Solution B (Light Salt)	Solution C (Conc. Salt)
Liquid Solution	100% Pure Tap / Distilled Water	Water + 0.5 tbsp Salt	Water + 2 tbsp Salt
Initial Potato Core	Identical ~50 mm length	Identical ~50 mm length	Identical ~50 mm length
Soaking Time	30 – 45 Minutes	30 – 45 Minutes	30 – 45 Minutes
Key Equipment	Ruler: Length measurement in mm Digital Balance: Mass measurement in grams (g) Paper Towel: Blotting surface moisture		
Expected Final Texture	Stiff, hard, turgid (firm)	Slightly flexible / neutral	Soft, limp, droopy (flaccid)

3. QUANTITATIVE DATA COLLECTION PROTOCOL & RECORD SHEET

Step-by-Step Laboratory Protocol:

1. Cut 3 uniform potato strips. Record initial texture.
2. Use ruler to measure initial length (L_1 in mm) and balance to measure initial mass (M_1 in grams).
3. Submerge one strip into each solution for 30–45 mins.
4. Remove strips, blot gently with paper towels, remeasure length (L_2) and mass (M_2).
5. Calculate Mass Change: $\Delta M = M_2 - M_1$ and Percentage Change: $\% \text{ Change} = (\Delta M \div M_1) \times 100$.

Q1. Student Laboratory Observations Table

Solution Type	Initial Length (L_1 mm)	Final Length (L_2 mm)	Initial Mass (M_1 g)	Final Mass (M_2 g)	Change in Mass (ΔM g)	Final Texture
Solution A (Pure Water)						
Solution B (Light Salt)						
Solution C (Conc. Salt)						

4. STUDENT PRACTICE & ASSESSMENT QUESTIONS

Q2. Multiple Choice Questions

a) Which tool is used to record the mass change of potato strips before and after soaking?

☐ (A) Measuring tape

☐ (B) Digital Balance

☐ (C) Thermometer

☐ (D) Graduated cylinder

b) During osmosis, water molecules naturally migrate:

☐ (A) From higher water concentration to lower water concentration

☐ (B) From lower water concentration to higher water concentration

☐ (C) From a dry area to an even drier area

☐ (D) Only through plant roots, never through potato cell walls

Q3. Fill in the Blanks (Vocabulary Check)

Use the word bank: [**Turgid** | **Semi-permeable** | **Shrink** | **Osmosis** | **Balance**]

1. A cell membrane that lets water pass through but blocks salt is called _____.

2. In Solution A (Pure Water), potato cells absorb water and become firm or _____.

3. In Solution C (Concentrated Salt), water leaves the cells, causing the potato to _____ and lose mass.

4. To measure precise weight changes in grams, scientists use a digital _____.

Q4. Experimental Analysis & Reasoning

a) Why is it important to blot the potato strips with paper towels before measuring final mass on the balance?

b) Explain why using rulers and digital balances provides better scientific evidence than just feeling texture with your hands:

Q5. Reversibility Challenge Prediction

Predict what would happen to the mass and length if a limp potato strip from Solution C is transferred back into Solution A (Pure Water) for 45 minutes. Explain the biological reason using the concept of osmosis:

Q6. Real-World Application: The Salted Salad Mystery

A chef salts a fresh green salad 2 hours before dinner. By serving time, the bowl is full of liquid and the leaves are soggy. Based on what you learned about osmosis, explain where that water came from: